MICROCONTROLLER-BASED VOLTAGE PROTECTION SYSTEM

1. Introduction and Objective

1.1 Introduction

This project is make for a Voltage Protection System using Microcontroller. The main work of this system is check the input voltage and protect the electrical device from high or low voltage. I am making this project to understand how voltage protection system works in real life. In this project I am using a voltage sensing circuit for measure the input voltage. Arduino Uno compare the voltage with a fixed limit and if the voltage become too high or too low, the relay disconnect the load automatically. A 20×4 LCD is use to show the voltage value and system status. I also use a fault LED for showing warning when any voltage problem is happen. The system information can be monitor on computer using serial communication. I am making and testing this circuit in Proteus software to see how it works in different voltage condition.

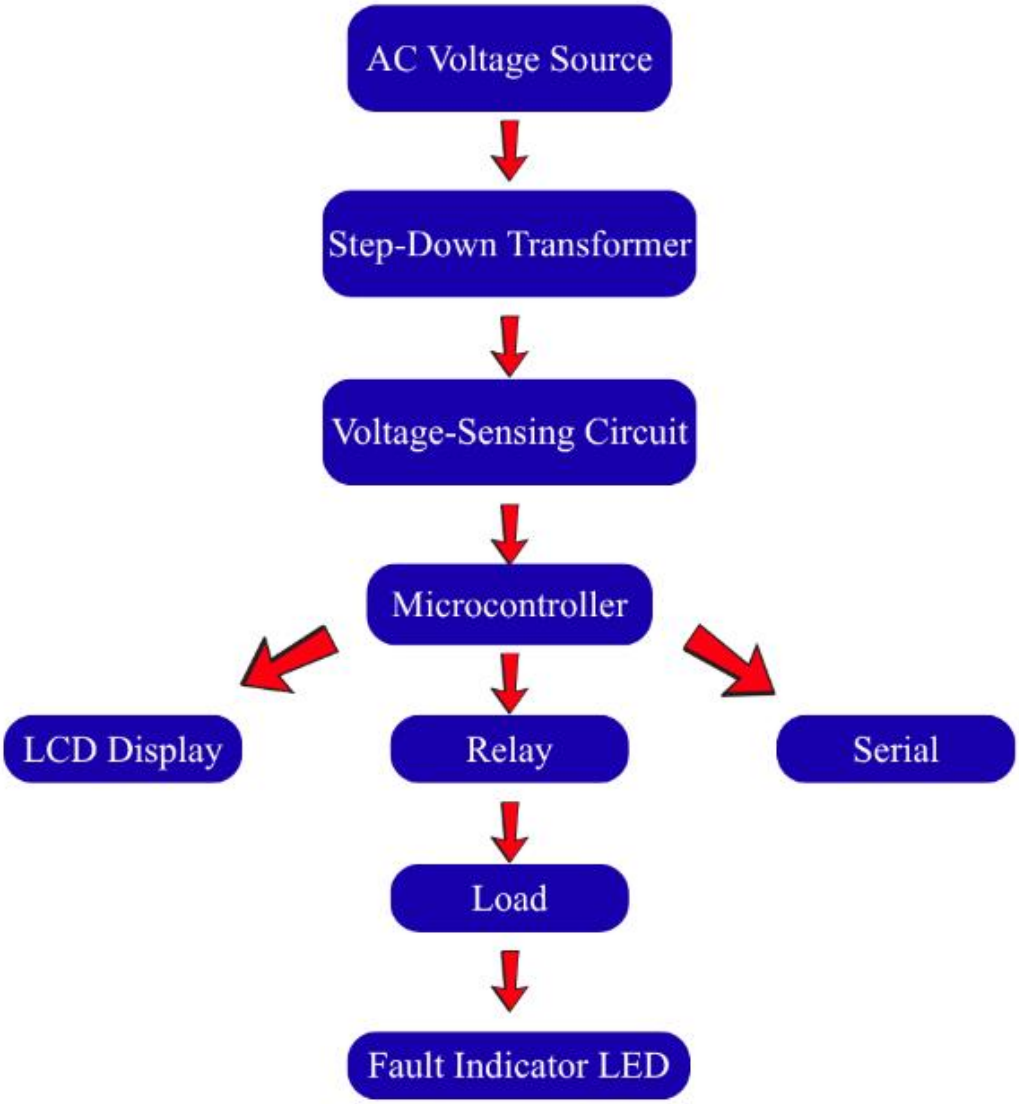
1.2 Objectives

The main objectives of this project is:

- To check the electrical supply voltage continuously.
- To find out high voltage and low voltage condition.
- To disconnect the load automatic when voltage become unsafe.
- To show the voltage value and protection status in LCD display.
- To use a LED for showing fault indication.
- To monitor the system information by using serial communication.

2. Block Diagram

The main functional blocks of the system are shown below.



3. Circuit Diagram and Component List

3.1 Circuit Diagram

The circuit was designed in Proteus using an Arduino Uno, a transformer-based voltage-sensing network, an LCD, an LED, and a relay connected to a simulated load.

The major connections are:

Component	Arduino Pin
Voltage sensor output	A0
LCD RS	D13
LCD Enable	D12
LCD D4	D11
LCD D5	D10
LCD D6	D9
LCD D7	D8
Relay control	D7
Fault indicator LED	D6
Serial TX	D1
Circuit ground	GND

3.2 Component List

Component	Quantity
Arduino Uno	1
Step-down transformer	1
Resistors	4
Capacitor, 10 µF	1
20×4 LCD	1
5 V relay	1
LED	1
AC voltage source	1
Virtual Terminal	1
Simulated electrical load	1

4. Voltage-Sensing Method

The system use a transformer based sensing circuit for measure the input AC voltage. The transformer reduce the voltage to a lower AC voltage. After that a resistor network is use to make the voltage suitable for Arduino Uno analog input. Arduino Uno have a 10-bit ADC which give digital reading from 0 to 1023 for input voltage around 0 to 5V. The program use EmonLib library for calculate the RMS voltage. In the program calibration value is set to 187 and phase shift value is 1.7. These value may need to change for get more accurate result according to the actual circuit. RMS voltage is important because AC voltage keep changing with time. It help to know the effective voltage value of the AC supply. This value is equal to the DC voltage which can produce same heating effect in a resistor.

5. Overvoltage and Undervoltage Protection Logic

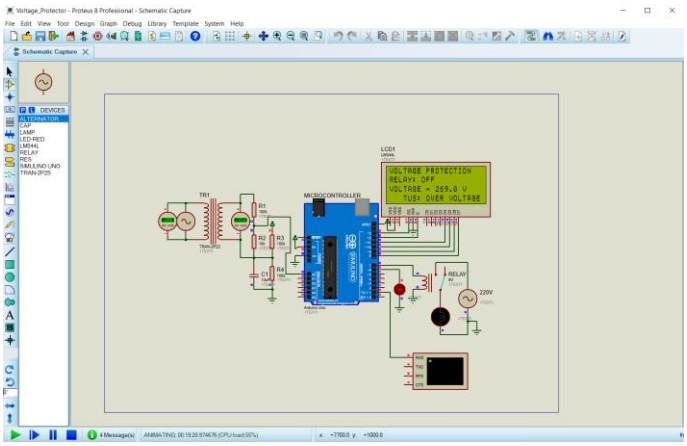
The Arduino continuously check the input voltage and compare it with two fixed voltage limits.

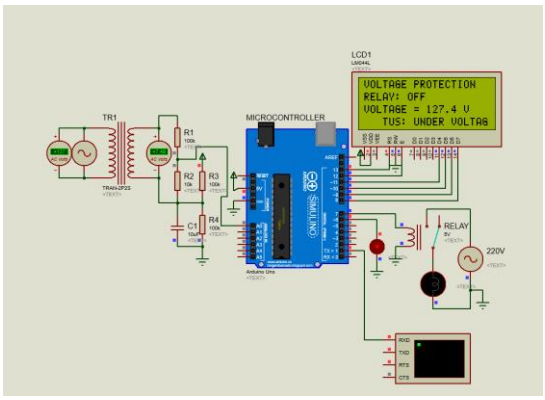
- **Lower voltage limit:** 180 V
- **Upper voltage limit:** 250 V

Measured Voltage	Condition	Relay	LED
Below 180 V	Undervoltage	OFF	ON
180–250 V	Normal voltage	ON	OFF
Above 250 V	Overvoltage	OFF	ON

7. Proteus Simulation Results

The circuit was simulated in Proteus to examine the voltage-protection logic, LCD output, and relay response.





An observed simulation screenshot showed the following:

Parameter	Observed Value
Displayed voltage	268.0 V
Protection status	OVER VOLTAGE
Relay status	OFF
Fault indicator	ON, according to the program logic

The displayed value of 268.0 V is above the configured upper threshold of 250 V. Therefore, the controller identifies an overvoltage condition and commands the relay to disconnect the load.

8.2 Future Improvements

The project can be improve in future by adding some new features:

- A current sensor like PZEM-004t can be use for measure the current.
- Overcurrent protection can be added for protect the load from high current.
- A recovery delay can be added so the system wait some time before connect the load again.
- Voltage measurement can be improve by proper sensor calibration.
- LCD display can show more details about voltage and fault condition.
- Voltage and current limit can be adjustable according to user need.
- Better sensing circuit and relay driver can be use for improve safety and performance.
- Watchdog and safe startup system can be added for make the system more reliable.

9. Conclusion

In this project, I am making a voltage protection system using Arduino Uno and Proteus software. The main purpose of this project is to protect electrical devices from high and low voltage. The system check the input voltage and show the voltage value on LCD. When the voltage goes above or below the fixed limit, the relay disconnect the load automatically.

During simulation, the voltage shows 268.0V which is higher than the limit of 250V. For this reason, the system detect the overvoltage condition and disconnect the load.